

SkillPath AI
AI-Based Career Guidance and Resume Skill
Analysis System

A Minor Project Report

Submitted in partial fulfillment of requirement of the
Degree of

BACHELOR OF TECHNOLOGY in COMPUTER
SCIENCE & ENGINEERING

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Report Approval

The project work “**SkillPath AI**” is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn therein; but approve the “Project Report” only for the purpose for which it has been submitted.

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Declaration

We hereby declare that the project entitled “ **SkillPath AI** ” submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in ‘B.Tech’ completed under the supervision of Prof. Mitushi Soni ,Dr. Mahesh Patidar, Faculty of Engineering, Medi-Caps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

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Certificate

I/We, Prof. Mitushi Soni ,Dr. Mahesh Patidar certify that the project entitled “ **SkillPath AI** ” submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by Deeksha Gupta,Deshna shivhare,Charvi sharma is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

SkillPath AI is an intelligent web-based system developed to provide career guidance and job readiness prediction for students and job seekers. In today's fast-growing digital world, many individuals face confusion while choosing the right career path due to lack of proper guidance and awareness about required skills. This project aims to solve this problem using Artificial Intelligence (AI) and Machine Learning (ML) techniques.

The system allows users to upload their resumes, which are analyzed using Natural Language Processing (NLP) to extract relevant skills. These skills are then compared with predefined datasets that contain information about different job roles and their required skills. Based on this comparison, the system predicts suitable career options such as Web Developer, Data Scientist, or AI Engineer.

In addition to role prediction, the system performs skill gap analysis to identify missing skills required for a particular job. It also provides personalized career roadmaps, including recommended courses, certifications, and projects to help users improve their skills. Furthermore, the system predicts job readiness score and selection probability using machine learning models.

The project is implemented using HTML, CSS, and JavaScript for the frontend, and Python (Flask) with MySQL for backend processing. The system ensures fast performance, user-friendly interface, and reliable results.

SkillPath AI helps users understand their strengths, improve their weaknesses, and make better career decisions. It reduces dependency on manual career counseling and provides a smart, data-driven solution. Overall, the system acts as a virtual career mentor and helps bridge the gap between academic learning and industry requirements.

This project helps users:

- Understand their skill level
- Identify missing skills
- Improve employability
- Make informed career decisions

Thus, SkillPath AI acts as a virtual career mentor and bridges the gap between academic learning and industry requirements.

Keywords:

Artificial Intelligence, NLP, Machine Learning, Resume Analysis, Career Recommendation, Skill Gap Analy

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Abbreviations

These are short forms used in project

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
UI	User Interface
DBMS	Database Management System
API	Application Programming Interface
TF-IDF	Term Frequency–Inverse Document Frequency
PDF	Portable Document Format
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript

Notations & Symbols

<u>Symbol / Notation</u>	<u>Meaning</u>
(x)	Input data (resume text provided by user)
(y)	Output (predicted job role)
(f(x))	Prediction function used in system
(P)	Probability (job selection probability)
(%)	Percentage (skill match, readiness score)
(n)	Number of data entries / skills
(TF)	Term Frequency (word occurrence in resume)
(IDF)	Inverse Document Frequency
(TF-IDF)	Technique used for skill extraction
(μ)	Mean value (used in ML calculations)
→	Process flow direction
↔	Two-way communication between modules
✓	Successful operation
✗	Error or failure
○ (Oval)	Start/End in flowchart

□ (Rectangle)	Process step
◇ (Diamond)	Decision making
→	Flow line
▭ (Parallelogram)	Input/Output operation
Entity	Database table
Attribute	Column in table
Relationship	Link between tables

Chapter-1

INTRODUCTION

1.1 Introduction

In the modern digital era, choosing the right career path has become a major challenge for students and job seekers. Many individuals lack proper guidance and are often unaware of the skills required for different job roles in the industry. As a result, they face confusion, poor career decisions, and difficulty in securing employment.

With the advancement of Artificial Intelligence (AI), it is now possible to build intelligent systems that can analyze user data and provide personalized recommendations. Resume analysis is one such application where a system can evaluate a user's skills and suggest suitable career paths.

The **SkillPath AI** project is developed as an AI-based career guidance and resume analysis system. It allows users to upload their resumes, extracts relevant skills, and matches them with predefined datasets to predict the most suitable job roles.

The system uses structured datasets (such as Kaggle datasets) that contain mappings between job roles and required skills. Based on this, it provides career recommendations such as Frontend Developer, Backend Developer, Data Scientist, or AI Engineer.

The main aim of the system is to help users:

- Understand their current skill set
- Identify missing skills
- Choose the right career path
- Improve their employability

The project is implemented using web technologies such as HTML, CSS, JavaScript for frontend and Python (Flask) with MySQL for backend operations

1.2 Literature Review

The rapid evolution of Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) has significantly transformed recruitment systems, recommendation engines, and intelligent decision-support platforms. With the growing emphasis on skill-based hiring and data-driven evaluation of candidates, automated resume analysis and career recommendation systems have become an active area of research. The development of SkillPath

AI is grounded in foundational work related to content-based recommendation systems, resume classification techniques, NLP-based text processing, and machine learning implementation frameworks.

One of the core components of SkillPath AI is its personalized roadmap and course recommendation module. The theoretical basis for such recommendation systems is well established in the literature. Lops, De Gemmis, and Semeraro [1] provide an extensive overview of content-based recommender systems, describing how user profiles and item profiles are represented and compared using similarity functions. According to [1], content-based systems operate by analyzing features of items and matching them with user preferences represented in vector space models. These systems rely heavily on feature extraction and weighting schemes such as TF-IDF to determine relevance scores. In the context of SkillPath AI, this concept is applied by representing user skills as a profile vector and mapping them against structured course metadata. By identifying missing skills and computing similarity between user profiles and course features, the system generates targeted recommendations. Furthermore, [1] emphasizes the importance of explainability in recommendation systems, ensuring that users understand why specific suggestions are made. This principle is incorporated into SkillPath AI by linking each recommended course or project directly to identified skill gaps.

In addition to recommendation systems, automated resume analysis has been widely explored in research. Malik and Kumar [2] investigate resume classification using machine learning techniques, demonstrating how unstructured resume data can be processed, structured, and classified into relevant job domains. Their methodology includes text preprocessing steps such as tokenization and stop-word removal, followed by feature extraction using TF-IDF. Various supervised learning algorithms are then applied to categorize resumes effectively. The work in [2] validates the feasibility of converting free-text resumes into structured feature representations suitable for machine learning models. This directly supports the resume parsing and skill extraction pipeline implemented in SkillPath AI. The classification framework discussed in [2] also demonstrates how predictive models can assist in job-role mapping, which forms a critical step in skill gap analysis. Additionally, the study highlights the importance of labeled training data and performance metrics such as precision, recall, and F1-score for evaluating model accuracy. These evaluation strategies are adopted in SkillPath AI to ensure reliability of classification and prediction outputs.

The theoretical underpinnings of NLP techniques used in resume analysis are extensively described in the foundational textbook by Jurafsky and Martin [3]. This work provides a comprehensive treatment of linguistic processing, statistical language models, and information retrieval techniques. Concepts such as tokenization, part-of-speech tagging, stemming, lemmatization, vector space modeling, and TF-IDF weighting are elaborated with mathematical formulations and practical examples. In SkillPath AI, these concepts are applied during the preprocessing stage of resume analysis. The vector space model described in [3] enables

representation of textual data as numerical vectors, allowing similarity computation between candidate profiles and job requirements. Moreover, the discussion of evaluation methodologies in [3] provides guidance on assessing system accuracy and robustness. The book also discusses semantic representation techniques, which inspire potential future improvements in SkillPath AI, such as integrating word embeddings or contextual models to enhance semantic understanding of skills.

While theoretical models provide the foundation, practical implementation of machine learning systems requires robust software frameworks. The Scikit-learn documentation [4] offers standardized and well-documented tools for implementing machine learning algorithms in Python. It provides utilities for feature extraction (e.g., TfidfVectorizer), model selection, classification, regression, and evaluation. In the proposed system, Scikit-learn is used to implement the job selection probability prediction module. Algorithms such as Logistic Regression and Random Forest are employed to estimate the likelihood of selection based on features such as skill match percentage, assessment score, experience level, and project count. The documentation in [4] outlines best practices for model validation, including cross-validation and hyperparameter tuning. These practices ensure that the predictive component of SkillPath AI maintains reliability and avoids overfitting. Additionally, Scikit-learn's modular pipeline architecture supports integration of preprocessing and classification steps, aligning with the modular design approach adopted in the system.

When synthesizing insights from these references, it becomes evident that the integration of NLP, recommender systems, and predictive modeling is essential for building an intelligent career guidance platform. Reference [1] provides the structural basis for personalized recommendations; reference [2] validates resume classification using machine learning; reference [3] offers the theoretical framework for NLP and text vectorization; and reference [4] supplies the practical implementation tools required for predictive modeling. Together, these works create a comprehensive methodological foundation for SkillPath AI.

Furthermore, the reviewed literature highlights certain challenges. Resume text is inherently unstructured and varies widely in formatting, which complicates extraction accuracy. Feature engineering and preprocessing therefore play a critical role in system performance, as emphasized in [2] and [3]. Another key challenge lies in generating meaningful probability predictions for job selection, since real-world hiring decisions depend on multiple external factors. The predictive modeling approach implemented in SkillPath AI addresses this by using interpretable ML models and clearly presenting probability outputs as estimations rather than guarantees. The explainability principle emphasized in [1] is also applied here, ensuring that predictions are accompanied by understandable reasoning.

In conclusion, the literature provides strong theoretical and practical justification for the design of SkillPath AI. The integration of content-based recommendation strategies [1], resume

classification methodologies [2], NLP foundations [3], and machine learning implementation tools [4] enables the development of a unified, intelligent, and scalable career guidance system. By building upon established research while addressing identified limitations, SkillPath AI contributes to the growing domain of AI-driven educational and recruitment support systems.

1.3 Objectives

The main objectives of the project are:

- To develop an AI-based career guidance system
- To analyze resumes and extract relevant skills
- To match user skills with job role datasets
- To predict suitable career roles
- To identify missing skills for improvement
- To provide a user-friendly web interface
- To reduce dependency on manual career counseling

1.4 Significance

The project is significant because it addresses a real-world problem faced by students and job seekers. Many individuals lack clarity about their career path and required skills.

SkillPath AI helps users make better decisions by providing data-driven recommendations. It improves career planning and increases chances of employment.

The system is also useful for educational institutions, as it helps students understand industry requirements and focus on skill development.

1.5 Research Design

The research design of the SkillPath AI project follows a problem-solving and system development approach.

The first phase involves requirement analysis, where the major problems such as lack of personalized guidance, skill mismatch, and resume inefficiency are studied.

The second phase focuses on data collection using Kaggle datasets containing job roles and their associated skills.

The third phase includes system design, where database structure, interface design, and system architecture are planned.

The fourth phase is implementation, where the frontend is developed using HTML, CSS, and JavaScript, while the backend uses Python Flask and MySQL for database management.

The fifth phase includes testing and validation to ensure correct login, resume upload, skill extraction, and role prediction.

Finally, result analysis is performed to evaluate system performance, usability, and prediction effectiveness.

This structured research design ensures systematic development and reliable outcomes for the project.

1.6 Source of Data

Primary Data

- User details (name, email)
- Resume content uploaded by users

Secondary Data

- Kaggle datasets for job roles and skills
- Research papers and online resources
- Documentation for Flask, MySQL, etc.

1.7 Chapter Scheme

The project report of **SkillPath AI** is divided into different chapters to present the work in a systematic way.

Chapter 1: Introduction

This chapter gives an overview of the project, including objectives, significance, literature review, and basic idea of the system.

Chapter 2: Requirement Specification

This chapter explains system requirements such as user characteristics, functional requirements, performance needs, and constraints.

Chapter 3: Design

This chapter describes system design, algorithms, diagrams (DFD, ER, etc.), and database structure.

Chapter 4: Implementation, Testing, and Maintenance

This chapter explains technologies used, system implementation, testing methods, and user instructions.

Chapter 5: Results and Discussion

This chapter shows system outputs, interface screens, and performance analysis.

Chapter 6: Summary and Conclusion

This chapter summarizes the project and presents final conclusions.

Chapter 7: Future Scope

This chapter discusses possible improvements and future enhancements of the system.

Chapter 2

REQUIREMENTS SPECIFICATION

2.1 User Characteristics

The system is designed for:

Students

- Basic computer knowledge
- Need career guidance
- Preparing for placements

Fresh Graduates

- Searching for jobs
- Need resume improvement
- Want skill-based suggestions

Professionals

- Looking for career switching
- Want skill gap analysis

Admin (Optional)

- Manages system data
- Updates datasets

2.2 Functional Requirements

The system must perform the following functions:

User Registration

- Allow new users to create accounts

User Login

- Authenticate users
- Display error for invalid credentials

Resume Upload

- Accept resume files
- Extract resume text

Skill Extraction

- Identify skills from resume

Role Prediction

- Match skills with dataset
- Predict suitable career role

Dashboard Display

- Show skills, role, and suggestions

Logout

- End user session securely

2.3 Dependencies

The project depends on:

Software

- Python
- Flask
- MySQL
- HTML, CSS, JavaScript

Tools

- VS Code
- MySQL Workbench
- Web Browser

Libraries

- pandas
- flask
- mysql connector

Dataset

- Kaggle dataset for skills and roles

2.4 Performance Requirements

- Fast response time (within seconds)
- Accurate skill matching
- Smooth UI interaction
- Reliable system performance
- Scalable for future expansion

2.5 Hardware Requirements

Minimum

- 4 GB RAM
- Intel i3 processor
- Internet connection

Recommended

- 8 GB RAM
- Intel i5 or higher
- SSD storage

2.6 Constraints and Assumptions

Constraints

- Limited dataset accuracy
- Resume format dependency
- Basic security implementation
- Internet required

Assumptions

- Users upload correct resumes
- Dataset contains valid skills
- System environment is properly configured

Chapter 3

DESIGN

3.1 Algorithm

The SkillPath AI system follows a step-by-step process for resume analysis and career prediction:

1. User logs into the system
2. Uploads resume (PDF/Text)
3. System extracts text from the resume
4. Extracted text is compared with skill dataset
5. Relevant skills are identified
6. Skills are matched with predefined job roles
7. Role with highest match score is selected
8. Predicted role and skills are displayed on dashboard

3.2 Function Oriented Design (Procedural Approach)

The system is divided into smaller functional modules:

- **Register Function** – Creates new user accounts
- **Login Function** – Authenticates users
- **Upload Function** – Handles resume upload
- **Skill Extraction Function** – Identifies skills from resume
- **Prediction Function** – Matches skills with roles
- **Dashboard Function** – Displays results

This modular design improves system organization and maintainability.

3.3 System Design

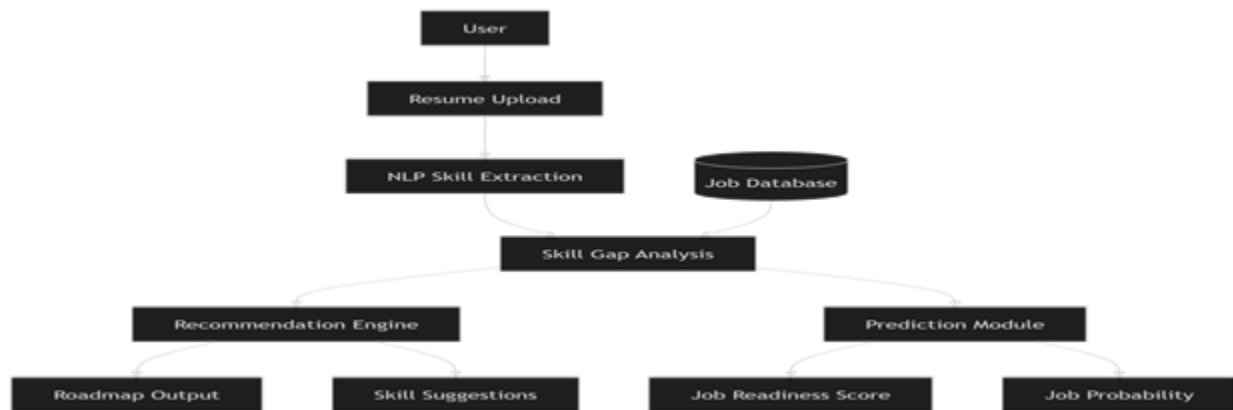
The system consists of:

- Frontend (HTML, CSS, JavaScript)
- Backend (Python Flask)
- Database (MySQL)
- Dataset (Kaggle for skill-role mapping)

It ensures smooth interaction between user input, processing, and output.

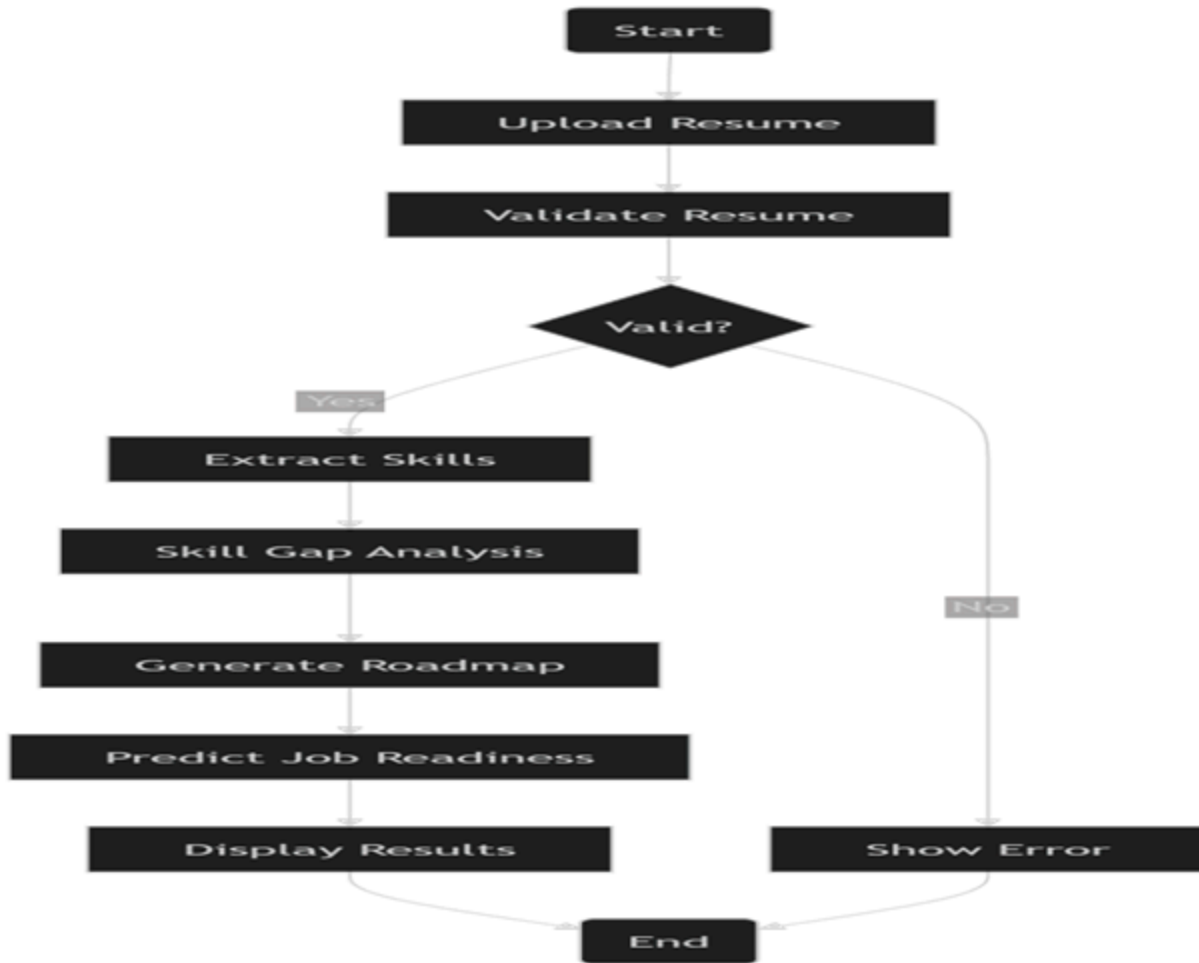
3.3.1 Data Flow Diagrams (DFD)

- In the **SkillPath AI** project, the DFD shows how the system takes the user's resume as input, processes it using NLP and machine learning, compares skills with job role requirements, and finally generates outputs like roadmap, skill suggestions, job readiness score, and job probability.
- In simple words, the DFD is the **working flow of information** inside the project



3.3.2 CONTROL FLOW DIAGRAM

In SkillPath AI, the Control Flow Diagram shows how the system starts from resume upload, validates the file, checks whether it is valid or not, and then moves through skill extraction, analysis, roadmap generation, prediction, and result display. If any problem occurs, the system shows an error and stops the process. This exact flow is also shown in the first PDF under the control flow diagram section



3.3.3 Flow Chart

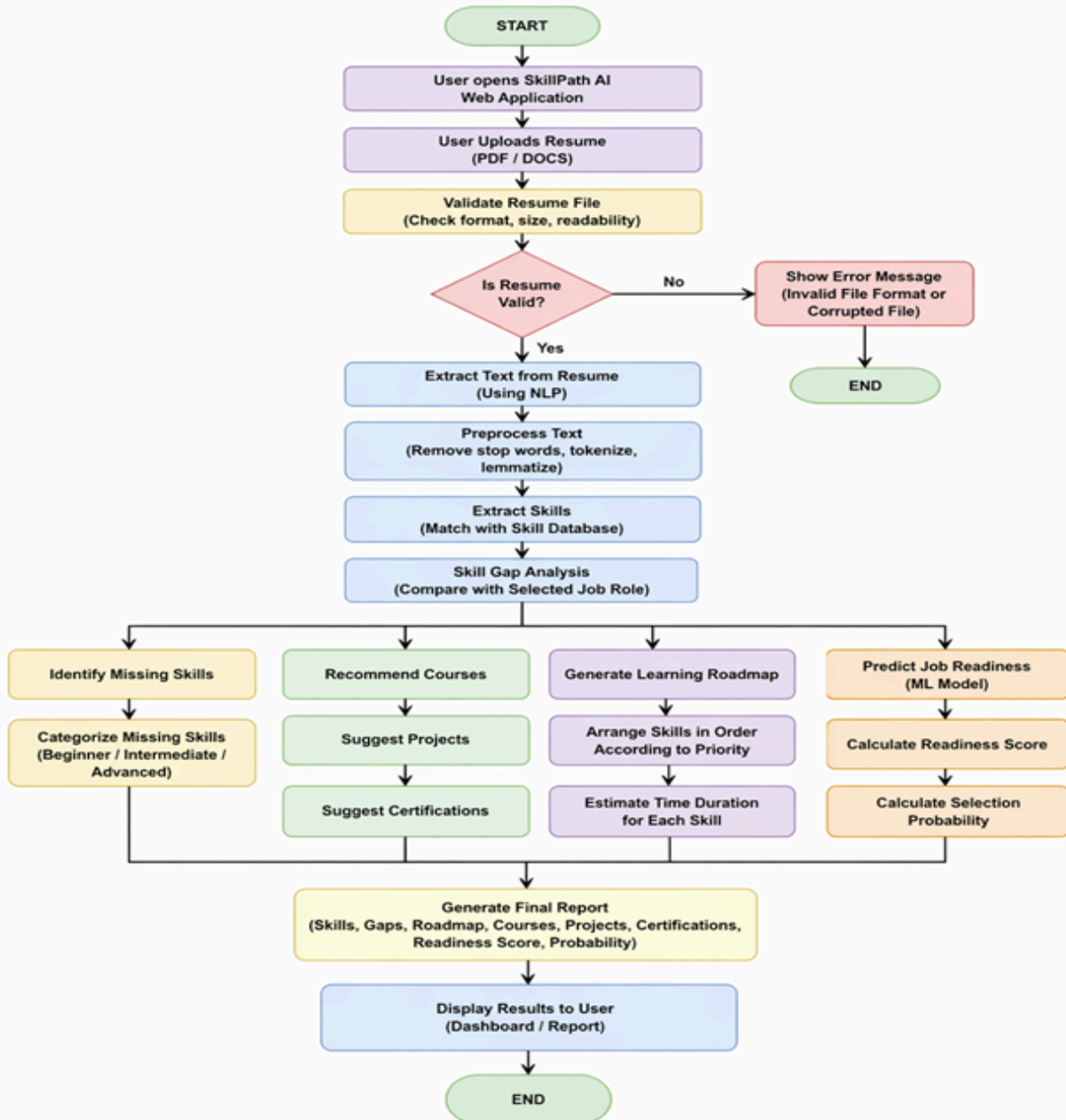
A flowchart is a diagram that shows the step-by-step working of a system. In the SkillPath AI project, it represents the complete process from user login to career prediction.

The process starts with user login, followed by validation of credentials. After successful login, the user uploads a resume. The system then extracts text, identifies skills, and matches them with a dataset to predict the most suitable career role. Finally, the results are displayed on the dashboard.

Thus, the flowchart helps in understanding the overall system process in a simple and clear manner.

3.3 COMPONENT DESIGN

3.3.1 FLOWCHART



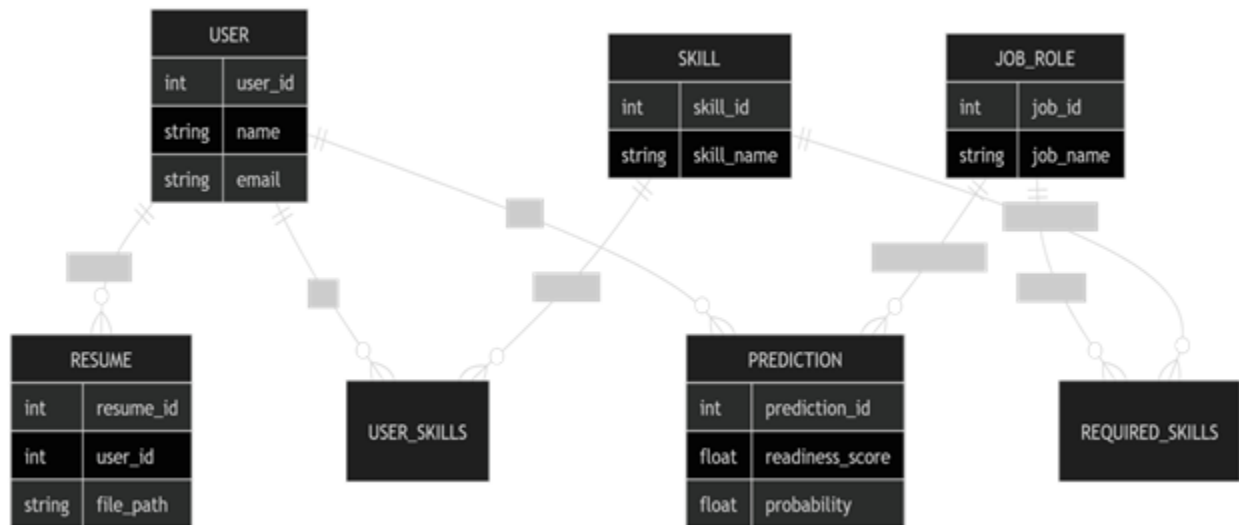
3.3.4 ER Diagram

The ER Model is used to show the database structure of the project. It tells us:

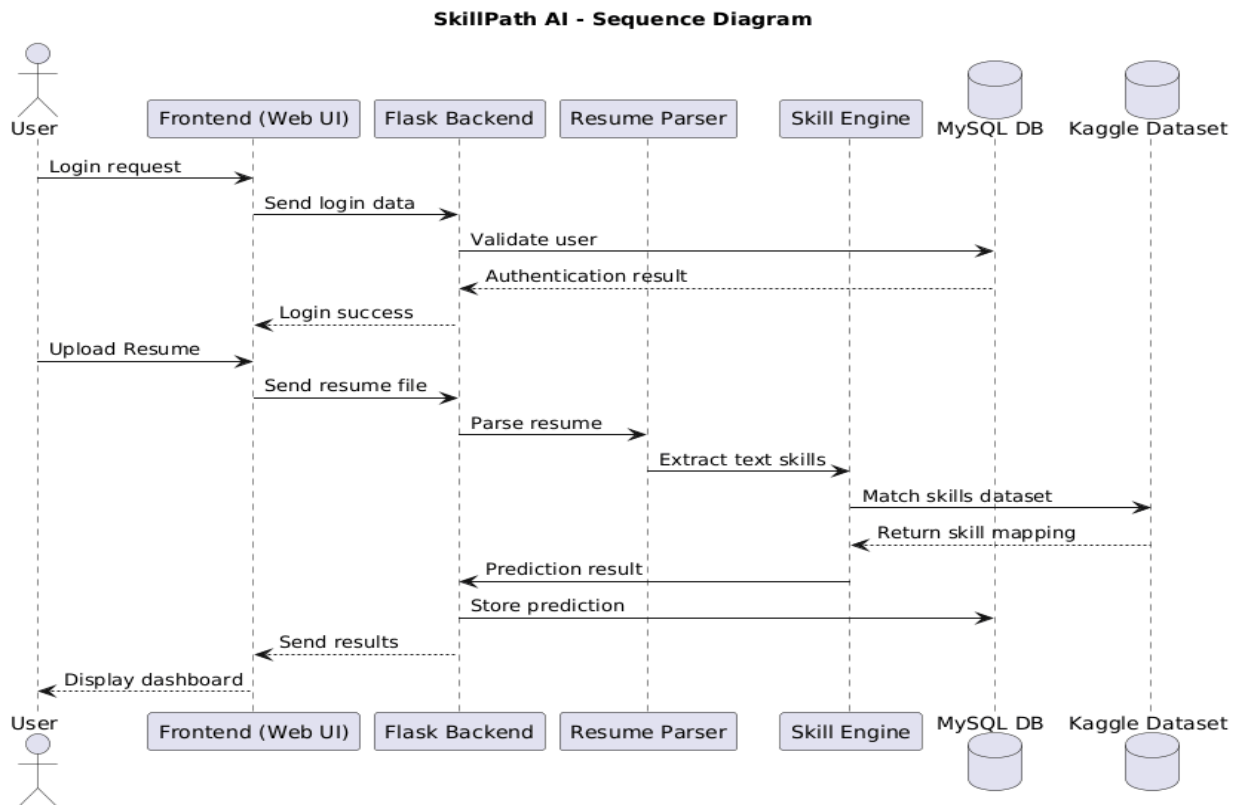
- what data we need to store,
- how that data is divided into tables/entities,
- and how one table is connected to another.

In SkillPath AI, the ER model is used to store user details, uploaded resumes, extracted skills, job roles, required skills, and prediction results.

This database design helps the system work properly because all modules like resume upload, skill extraction, skill gap analysis, roadmap generation, and prediction need organized data.



3.3.5 Sequence Diagram



In the SkillPath AI system, the sequence diagram illustrates the step-by-step interaction between the user, frontend interface, backend server (Flask), database, and prediction engine during the process of login, resume upload, skill extraction, and career prediction.

The diagram helps in understanding the dynamic behavior of the system and how data flows between different modules to generate the final output.

3.4 Database Design

Database design is a critical part of SkillPath AI because it stores user details, resumes, extracted skills, and prediction results efficiently.

The database is normalized up to 3NF to reduce redundancy and improve consist

3.4.1 Logical Database Design

Logical design defines entities, attributes, and relationships.

1. USERS Entity

Description:

Stores all user registration and login details.

Importance:

Used for authentication and session management.

2. RESUME Entity

Description:

Stores uploaded resume files and extracted text content.

Importance:

Forms the base for skill analysis and prediction.

3. SKILLS Entity

Description:

Stores all technical and professional skills.

Importance:

Used for skill extraction and matching.

4. USER_SKILLS Entity

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT	
id	INT	Unique ID	Primary Key, Auto Increment	
userID	INT	Reference to user	Foreign Key	
skillID	INT	Reference to skill	Foreign Key	

5. ROLES Entity

Description:

Stores career roles such as Frontend Developer, Backend Developer, Data Scientist, etc.

Importance:

Used for career role prediction.

6. ROLE_SKILLS Entity

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT	
id	INT	Unique ID	Primary Key	
roleID	INT	Reference to role	Foreign Key	
skillID	INT	 Required skill	Foreign Key	

7. PREDICTIONS Entity

Description:

Stores final prediction results and match scores.

Importance:

Maintains history of user career recommendations.

3.4.2 Physical Database Design-

Physical database design defines the actual implementation of the logical database structure in the database management system. It focuses on how the data is physically stored, accessed, and managed using SQL queries, table creation, constraints, indexing, and relationships. In the **SkillPath AI** project, the physical database is implemented using **MySQL** as the relational database management system. The database is designed to store user details, uploaded resumes, extracted skills, role mappings, and prediction results efficiently.

Database Used

MySQL

Main Tables Created

The following physical tables are created in the MySQL database:

1. USERS
2. RESUME
3. SKILLS
4. USER_SKILLS
5. ROLES
6. ROLE_SKILLS
7. PREDICTIONS

1. USERS Table -
CREATE TABLE Users (

```

    userID INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
2. RESUME Table
CREATE TABLE Resume (
    resumeID INT AUTO_INCREMENT PRIMARY KEY,
    userID INT,
    file_name VARCHAR(255) NOT NULL,
    content TEXT NOT NULL,
    uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (userID) REFERENCES Users(userID)
);
3. SKILLS Table
CREATE TABLE Skills (
    skillID INT AUTO_INCREMENT PRIMARY KEY,
    skill_name VARCHAR(100) UNIQUE NOT NULL
);
4. USER_SKILLS Table
CREATE TABLE User_Skills (
    id INT AUTO_INCREMENT PRIMARY KEY,
    userID INT,
    skillID INT,
    FOREIGN KEY (userID) REFERENCES Users(userID),
    FOREIGN KEY (skillID) REFERENCES Skills(skillID)
);
5. ROLES Table
CREATE TABLE Roles (
    roleID INT AUTO_INCREMENT PRIMARY KEY,
    role_name VARCHAR(100) NOT NULL
);
6. ROLE_SKILLS Table
CREATE TABLE Role_Skills (
    id INT AUTO_INCREMENT PRIMARY KEY,
    roleID INT,
    skillID INT,
    FOREIGN KEY (roleID) REFERENCES Roles(roleID),
    FOREIGN KEY (skillID) REFERENCES Skills(skillID)
);
7. PREDICTIONS Table
CREATE TABLE Predictions (
    predictionID INT AUTO_INCREMENT PRIMARY KEY,

```

```
userID INT,  
roleID INT,  
score FLOAT NOT NULL,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (userID) REFERENCES Users(userID),  
FOREIGN KEY (roleID) REFERENCES Roles(roleID)  
);
```

Chapter 4

IMPLEMENTATION, TESTING, AND MAINTENANCE

4.1 Introduction to Languages, IDE's, Tools and Technologies used for Implementation

The successful implementation of the SkillPath AI project requires the use of various programming languages, frameworks, development tools, and database technologies. These technologies work together to build an efficient AI-based career guidance and resume analysis system.

The project follows a web-based architecture where the frontend manages user interaction, the backend handles logic and processing, the database stores information, and the Kaggle dataset provides skill-role mapping for career prediction.

The major technologies used in the project are explained below.

Category	Technology / Tool	Purpose / Description
Programming Languages	Python	Backend development and resume processing
	JavaScript	Frontend interactivity and dynamic behavior
	SQL	Database queries and data management
Frontend Technologies	HTML	Structure of web pages
	CSS / Tailwind CSS	Styling and responsive design

Backend Technologies	Flask	Web framework for handling server-side logic
Database	MySQL	Stores user data, resumes, and prediction results
Dataset Source	Companies Website	Provides job roles and skill mapping for prediction
IDE / Tools	Visual Studio Code	Code development and debugging
	Postgres Workbench	Database design and management
	Web Browser (Chrome)	Application testing and execution
Testing Tools	Manual Testing	Verifying system functionality
Version Control	Git & GitHub	Code management and version tracking

4.2 Testing Techniques and Test Plans

Unit Testing:

Each module (login, resume upload, skill extraction, prediction) is tested individually to ensure correctness.

• Integration Testing:

Checks whether different modules work together properly (e.g., resume upload → skill extraction → prediction).

- **System Testing:**

Tests the entire application as a complete system to verify overall functionality.

- **User Interface Testing:**

Ensures that all pages, buttons, and forms are responsive, user-friendly, and error-free.

- **Validation Testing:**

Confirms that the system meets user requirements and produces expected outputs.

Test Plan:

Test Case ID	Module	Input	Expected Output	Result
TC001	Login	Valid username & password	User successfully logged in and redirected to dashboard	Pass
TC002	Login	Invalid username or password	Error message displayed	Pass
TC003	Resume Upload	Valid resume file (PDF/Text)	Resume uploaded successfully	Pass
TC004	Skill Extraction	Resume containing skills (Python, SQL)	Skills correctly extracted and displayed	Pass
TC005	Prediction	Extracted skills matched with dataset	Appropriate career role predicted	Pass

4.3 Installation Instructions

The following steps are required to install and run the SkillPath AI project successfully.

Step 1: Install Python

First, install Python on your system because the entire backend of SkillPath AI runs on Python. During installation, ensure “Add to PATH” is selected.

• Step 2: Install Required Libraries

Install necessary Python libraries using pip such as Flask (for web framework), Pandas (for data handling), and MySQL Connector (for database connection).

• Step 3: Install MySQL Database

Install MySQL Server and MySQL Workbench to create and manage the database used in the project.

• Step 4: Create Database and Tables

Create a database in MySQL and define required tables like Users, Resume, Skills, and Predictions to store project data.

• Step 5: Add Kaggle Dataset

Download the dataset from Kaggle and place it in the project folder. This dataset is used for mapping skills with job roles.

• Step 6: Organize Project Files

Arrange the project structure properly with folders such as templates (HTML files), static (CSS/JS files), and dataset folder.

• Step 7: Run the Application

Execute the main Python file (app.py) to start the Flask server and run the application locally.

• Step 8: Open in Browser

Open a web browser and access the system using the local address (<http://127.0.0.1:5000>) to use the application.

4.4 End User Instructions

The SkillPath AI system is designed to be simple and user-friendly. The following steps explain how end users can use the system effectively.

Step 1: Open Application

User opens the SkillPath AI web application in a browser.

• Step 2: Register/Login

New users register with details, while existing users log in using credentials.

• Step 3: Upload Resume

User uploads resume in PDF or text format for analysis.

• Step 4: Skill Extraction

System automatically reads the resume and extracts important skills.

• Step 5: Career Prediction

System compares skills with dataset and predicts suitable career role.

• Step 6: View Dashboard

User sees extracted skills, predicted role, and suggestions for improvement.

• Step 7: Analyze Results

User reviews skill gaps and career guidance provided by the system.

• Step 8: Logout

User logs out safely after completing the session.

Chapter 5

RESULTS AND DISCUSSIONS

5.1 User Interface Representation (of Respective Project)

The SkillPath AI system is designed with a modern, responsive, and user-friendly interface to ensure smooth interaction between users and the application. The main objective of the interface design is to provide easy accessibility, attractive visual representation, and efficient navigation throughout the system.

The frontend of the project is developed using HTML, CSS (Tailwind CSS), and JavaScript, while Flask is used to connect the frontend with backend operations and MySQL database management.

The interface includes several important pages that help users interact with the system efficiently.

1. Landing Page

The landing page acts as the homepage of the system and introduces the user to the SkillPath AI platform.

Features:

- Attractive and modern UI design
- Brief introduction to the project
- Navigation options for Login and Sign Up
- Responsive design for desktop and mobile users

Purpose:

It helps users understand the purpose and functionality of the project before using the system.

2. Login Page

The login page provides secure access to registered users.

Features:

- Username/Email field
- Password field

- Forgot Password option
- Sign Up redirection
- Error message for invalid credentials
- Modern glassmorphism and gradient design

Purpose:

It ensures secure authentication and user session management.

3. Registration Page

The registration page allows new users to create an account.

Features:

- Full name input
- Email input
- Password creation
- Validation checks
- Login redirection for existing users

Purpose:

It helps new users register before accessing system features.

4. Resume Upload Page

This is the core page of the project where users upload their resumes.

Features:

- Resume upload button
- File name display
- Resume parsing trigger
- Clean and simple UI

Purpose:

It starts the process of skill extraction and career prediction.

5. Dashboard Page

The dashboard displays the final results after analysis.

Features:

- Extracted skills list
- Predicted career role
- Match score
- Missing skills suggestions
- Career guidance recommendations

Purpose:

It provides users with AI-based career recommendations and skill improvement guidance.

6. Logout Interface

This feature allows users to safely exit the system.

Features:

- Session termination
- Redirect to login page

Purpose:

Ensures secure logout and user session control.

5.2 Brief Description of Various Modules of the System

The SkillPath AI project is divided into different modules to improve maintainability, performance, and scalability.

Module 1: User Authentication Module

This module manages user registration and login operations.

Functions:

- User registration
- Login validation
- Session management
- Logout functionality

Purpose:

To provide secure and personalized access to users.

Module 2: Resume Upload Module

This module handles the uploading and storage of resumes.

Functions:

- Resume upload
- File validation
- Resume text extraction
- Resume storage

Purpose:

To capture the professional profile of the user for analysis.

Module 3: Skill Extraction Module

This module processes the uploaded resume and identifies skills.

Functions:

- Resume parsing
- Keyword matching
- Skill detection
- Skill storage

Purpose:

To identify relevant technical and professional skills from the resume.

Module 4: Prediction Engine Module

This is the main AI-based recommendation engine.

Functions:

- Load Kaggle dataset
- Compare extracted skills with job roles
- Calculate matching score
- Predict suitable career role

Purpose:

To generate personalized career recommendations using skill-role mapping.

Module 5: Dashboard Module

This module displays final outputs to the user.

Functions:

- Show extracted skills
- Display predicted role
- Show missing skills
- Provide improvement suggestions

Purpose:

To help users understand their current skill profile and improve career planning.

Module 6: Database Management Module

This module handles all database operations.

Functions:

- Store user data
- Save resume details
- Maintain prediction history
- Manage skill-role records

Purpose:

To ensure structured and reliable data management.

5.3 Snapshots of System with Brief Detail of Each

This section contains the screenshots of the developed system along with brief explanations.

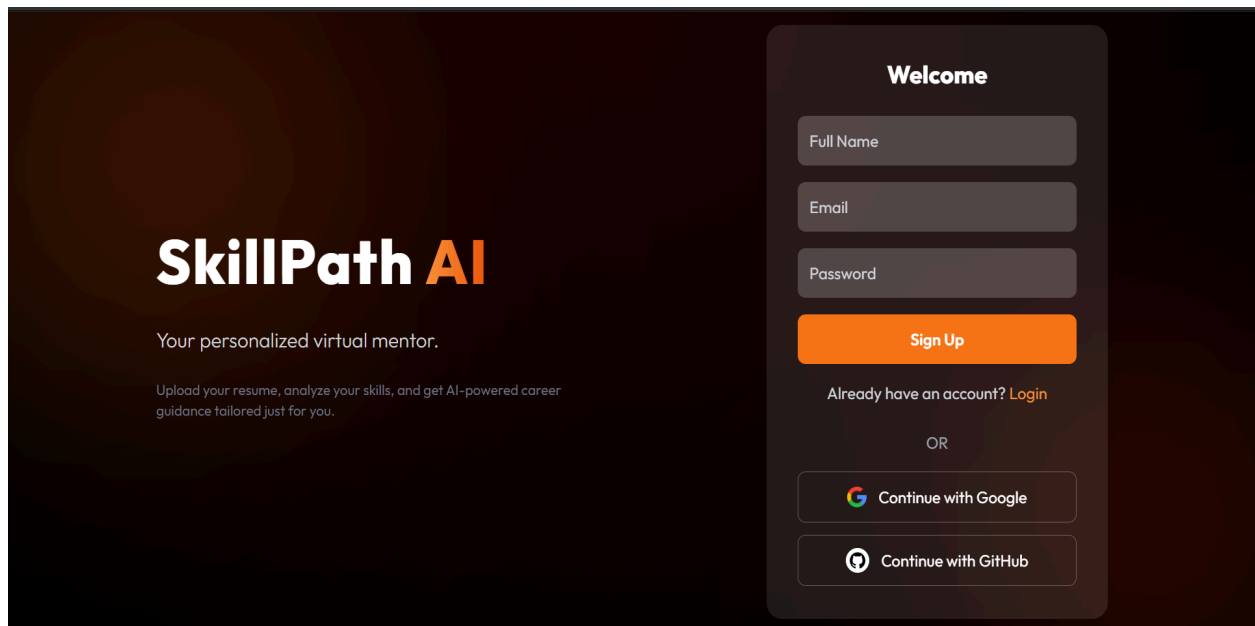
Snapshot 1: Login Page

Description:

This page allows registered users to log in using email and password.

Purpose:

Ensures secure access to the system.



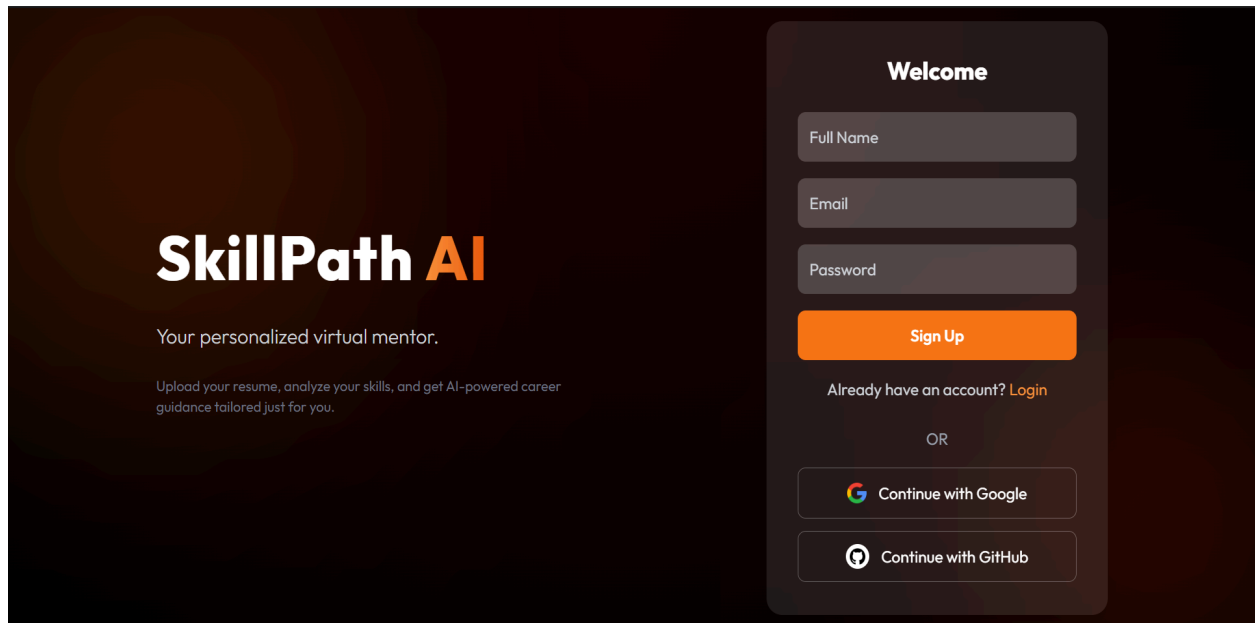
Snapshot 2: Registration Page

Description:

This page allows new users to create an account.

Purpose:

Used for new user onboarding.



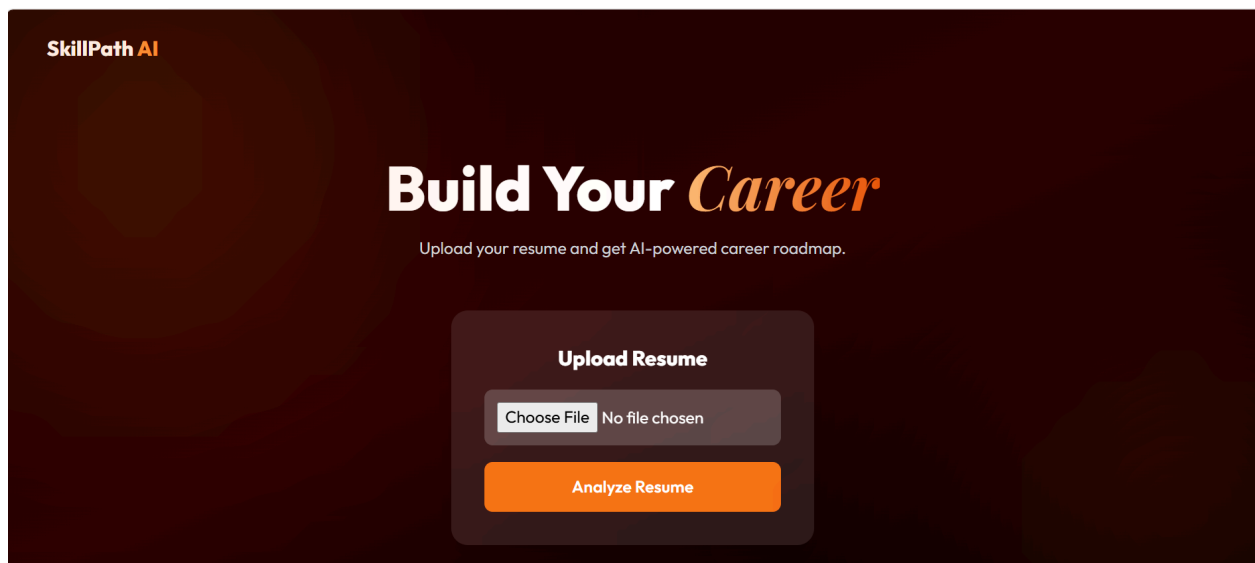
Snapshot 3: Resume Upload Page

Description:

Users upload resumes in PDF or text format using this page.

Purpose:

Starts the resume analysis process.



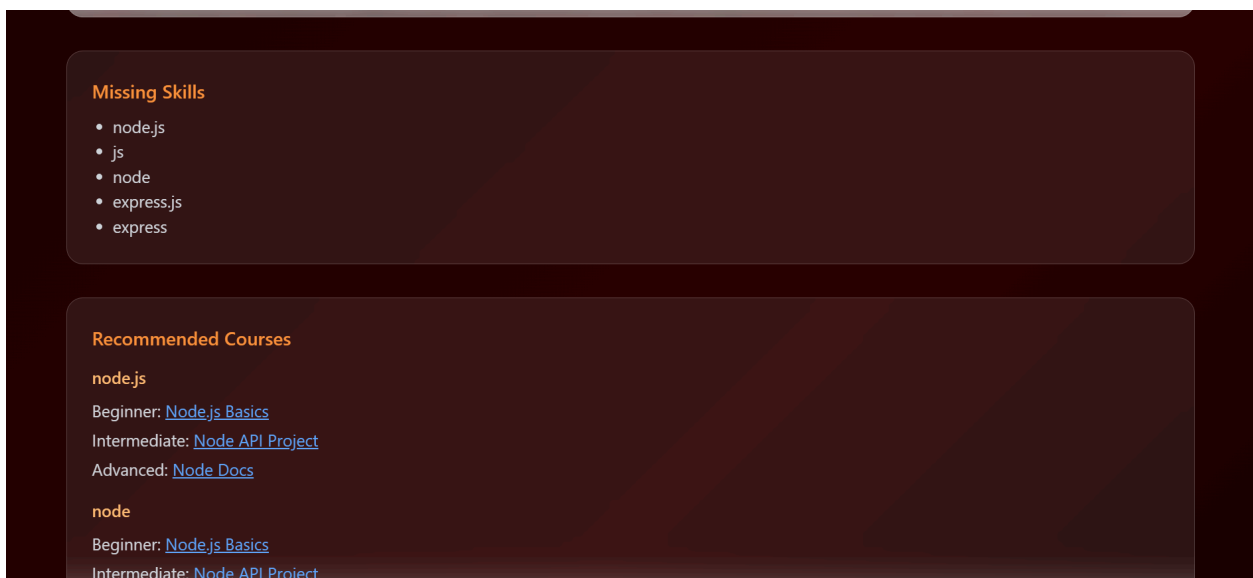
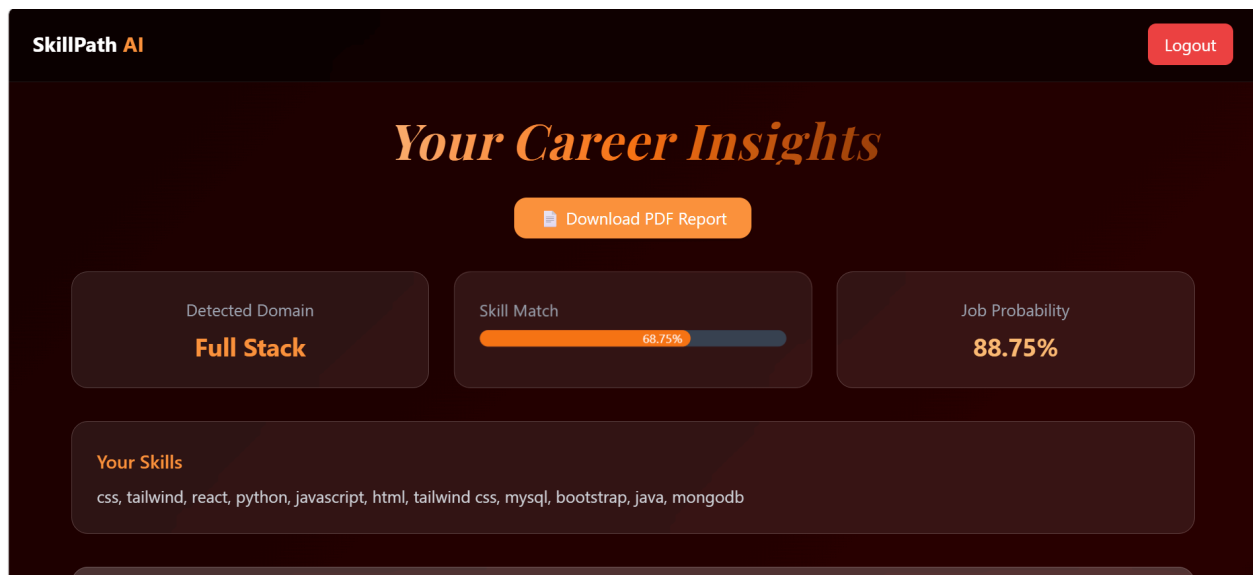
Snapshot 4: Dashboard Page

Description:

Displays extracted skills, predicted career role, and improvement suggestions.

Purpose:

Provides the main output of the system.



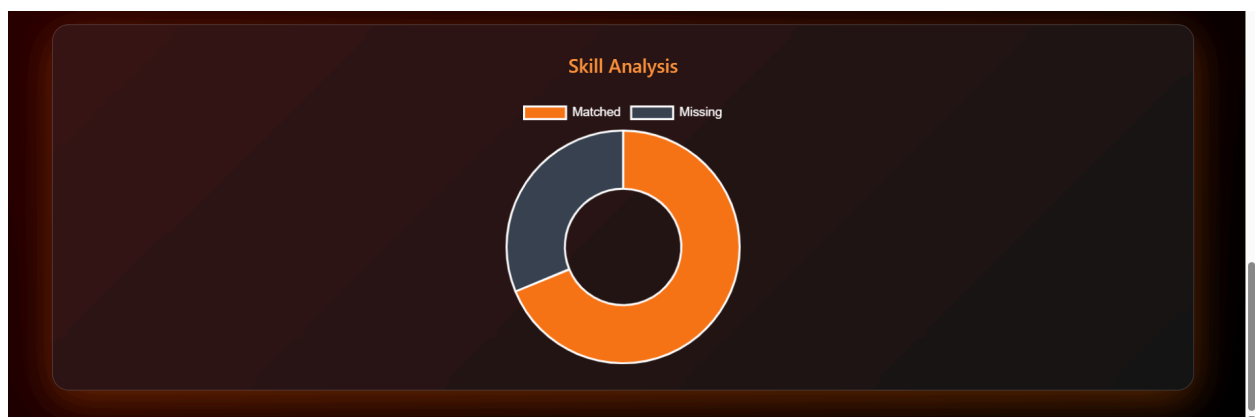
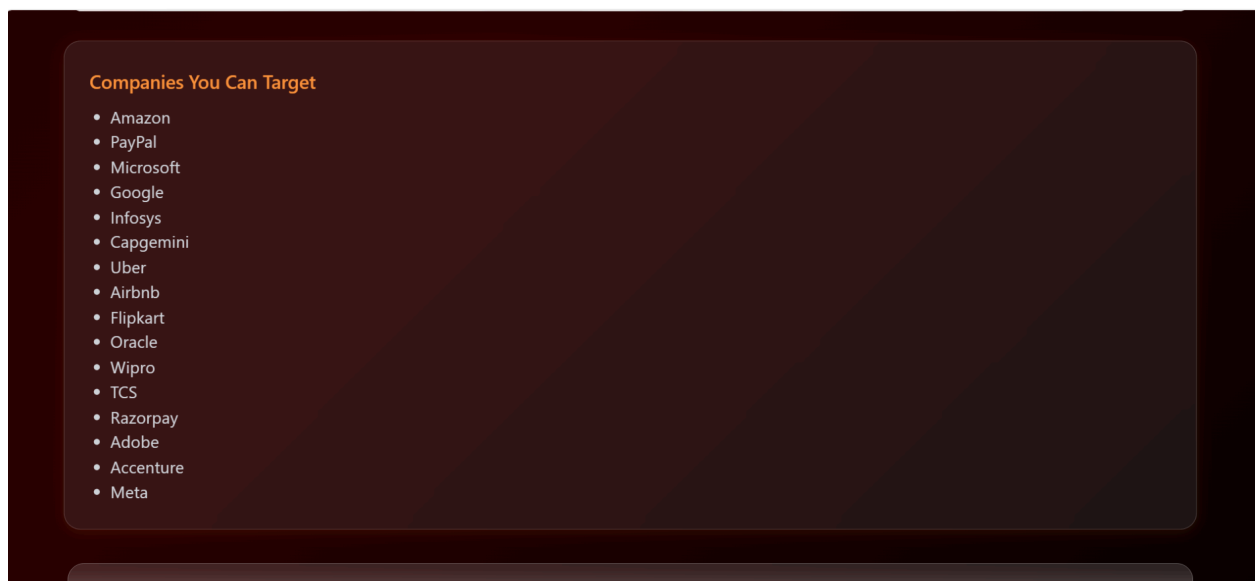
Snapshot 5: Final Prediction Result Page

Description:

Shows final matched career path and career recommendations.

Purpose:

Helps users understand their best-fit job domain.



5.4 Back Ends Representation (Database to be used)

The backend of SkillPath AI is developed using Python Flask with MySQL as the database management system.

The backend performs:

- User authentication
- Resume upload handling
- Skill extraction
- Role prediction
- Database operations
- Result storage

The backend ensures smooth communication between frontend UI and database.

Database Used: MySQL

MySQL is used because it is secure, reliable, and suitable for structured relational database systems.

It supports:

- Fast query execution
- Data consistency
- Foreign key relationships
- Normalization
- Scalability

Major Tables Used :

USERS Table

Stores user registration and login details.

Fields:

- userID
- name
- email
- password
- created_at

RESUME Table

Stores uploaded resume details and extracted content.

Fields:

- resumeID
- userID
- file_name
- content
- uploaded_at

SKILLS Table

Stores all available technical and professional skills.

Fields:

- skillID
- skill_name

USER_SKILLS Table

Stores mapping between users and extracted skills.

Fields:

- id
- userID
- skillID

ROLES Table

Stores predefined career roles.

Fields:

- roleID
- role_name

ROLE_SKILLS Table

Stores required skills for each job role.

Fields:

- id
- roleID
- skillID

PREDICTIONS Table

Stores final career prediction results.

Fields:

- predictionID
- userID
- roleID
- score
- created_at

5.5 Snapshots of Database Tables with Brief Description

This section includes screenshots of MySQL database tables created during implementation.

(Students should insert actual table screenshots from MySQL Workbench or phpMyAdmin here.)

Table Snapshot 1: USERS Table

Description:

Stores all user registration and login details.

Importance:

Used for authentication and session management.

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT	
userID	INT	Unique ID for user	Primary Key, Auto Increment	
name	VARCHAR(100)	Full name of user	Not Null	
email	VARCHAR(100)	User email address	Unique, Not Null	
password	VARCHAR(255)	User login password	Not Null	
created_at	TIMESTAMP	Account creation time	Default CURRENT_TIMESTAMP	

Table Snapshot 2: RESUME Table

Description:

Stores uploaded resume files and extracted text content.

Importance:

Forms the base for skill analysis and prediction.

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT
resumeID	INT	Unique ID for resume	Primary Key, Auto Increment
userID	INT	Reference to user	Foreign Key (Users.userID)
file_name	VARCHAR(255)	Resume file name	Not Null
content	TEXT	Extracted resume text	Not Null
uploaded_at	TIMESTAMP	Upload time	Default CURRENT_TIMESTAMP



Table Snapshot 3: SKILLS Table

Description:

Stores all technical and professional skills.

Importance:

Used for skill extraction and matching.

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT
skillID	INT	Unique skill ID	Primary Key, Auto Increment
skill_name	VARCHAR(100)	Name of skill	Unique, Not Null

Table Snapshot 4: ROLES Table

Description:

Stores career roles such as Frontend Developer, Backend Developer, Data Scientist, etc.

Importance:

Used for career role prediction.

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT	
roleID	INT	Unique role ID	Primary Key	
role_name	VARCHAR(100)	Career role name	Not Null	

Table Snapshot 5: PREDICTIONS Table

Description:

Stores final prediction results and match scores.

Importance:

Maintains history of user career recommendations.

FIELD NAME	DATATYPE	DESCRIPTION	CONSTRAINT
predictionID	INT	Unique ID	Primary Key
userID	INT	Reference to user	Foreign Key
roleID	INT	Predicted role	Foreign Key
score	FLOAT	Matching score	Not Null
created_at	TIMESTAMP	Prediction time	Default CURRENT_TIMESTAMP

Discussion of Results

The SkillPath AI system successfully performs resume analysis and predicts suitable career roles using skill-based matching logic.

The frontend design ensures smooth user interaction, while the backend logic provides reliable performance and data consistency.

The database structure supports scalability and future expansion, making the project suitable for real-world enhancement.

Overall, the system successfully achieves its objective of providing AI-powered career guidance and helping users make better career decisions based on their actual skills and professional profile.

Chapter 6

SUMMARY AND CONCLUSIONS

6.1 Summary

The project **SkillPath AI – AI-Based Career Guidance and Resume Skill Analysis System** was developed to provide an intelligent and user-friendly platform for students, fresh graduates, and job seekers to identify suitable career paths based on their skills and resume analysis.

In today's competitive job market, many individuals face difficulty in choosing the right career direction due to a lack of proper guidance and awareness of industry skill requirements. Traditional career counseling methods are often manual, time-consuming, and not personalized. To overcome these challenges, SkillPath AI provides an automated solution using Artificial Intelligence concepts and dataset-driven analysis.

The system allows users to register and log in securely, upload their resumes, and receive career recommendations based on the skills extracted from the resume content. The uploaded resume is analyzed using predefined datasets obtained from Kaggle, where job roles are mapped with required technical and professional skills.

The project uses technologies such as **HTML, CSS, JavaScript, Python (Flask), MySQL, and Kaggle datasets** for complete implementation. The frontend provides an interactive and responsive user interface, while the backend handles resume analysis, skill extraction, role prediction, and database operations.

The major modules implemented in the project include:

- User Authentication Module
- Resume Upload Module
- Skill Extraction Module
- Prediction Engine Module
- Dashboard Display Module
- Database Management Module

The database is designed using normalization up to Third Normal Form (3NF) to reduce redundancy and ensure data consistency. Multiple UML diagrams such as DFD, ER Diagram, Activity Diagram, Class Diagram, and Sequence Diagram were used for proper system design and documentation.

Testing was performed using unit testing, integration testing, and system testing to ensure correct functionality and reliable performance. The results showed that the system successfully predicts suitable career roles such as Frontend Developer, Backend Developer, Data Scientist, and AI Engineer based on user skills.

The project demonstrates how AI can be effectively applied in education and career development to bridge the gap between academic preparation and industry expectations.

6.2 Conclusions

The SkillPath AI project successfully achieves its main objective of providing an AI-based career guidance system through resume analysis and skill-role matching.

The system helps users understand their strengths, identify missing skills, and select the most suitable career path according to their technical profile. This improves confidence in career planning and supports better decision-making for placements, internships, and professional development.

The use of Kaggle datasets makes the project more practical and industry-relevant by providing real-world skill mappings for career prediction. Instead of relying only on manual counseling, the system provides fast, accurate, and scalable recommendations through automation.

The project also proves that even simple AI concepts such as keyword matching and dataset-driven prediction can create highly useful real-world applications. The structured database design, responsive interface, and modular architecture make the system easy to maintain and enhance in the future.

From an academic perspective, the project provides strong practical exposure in web development, database management, system design, and AI-based problem solving. It also helps students understand the importance of integrating frontend, backend, and intelligent recommendation systems into one complete solution.

Thus, SkillPath AI is not only a successful academic mini project but also a valuable real-world application that can be expanded into a full-scale professional platform for career counseling and recruitment support.

In conclusion, the project demonstrates that technology-driven career guidance systems can significantly improve employability, skill development, and professional growth for students and job seekers in the modern digital world.

Chapter 7

Future Scope

SkillPath AI has been successfully developed as an AI-based career guidance system. However, there are several areas where the system can be improved and expanded in the future.

- Integration with real-time job portals like LinkedIn and Naukri to provide live job opportunities
- Addition of AI chatbot for personalized career guidance and user interaction
- Implementation of interview preparation modules including mock interviews
- Enhancement of machine learning models for more accurate prediction results
- Support for multiple languages for better accessibility
- Development of a mobile application for Android and iOS users
- Integration of advanced analytics for tracking user progress and performance
- Use of deep learning techniques for better resume understanding

Thus, the system has strong potential for future enhancement and real-world deployment.

Appendix

Appendix A: Glossary

Term	Description
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Technique used for text analysis

Skill Gap Analysis Process of identifying missing skills

Job Readiness Score Measure of user's job preparedness

Prediction Model ML model used for career prediction

Appendix B: Sample SQL Tables

Example:

```
CREATE TABLE Users (  
    userID INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(100),  
    password VARCHAR(255)  
);
```

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